OENG1209: Engineering Science

Week 9:

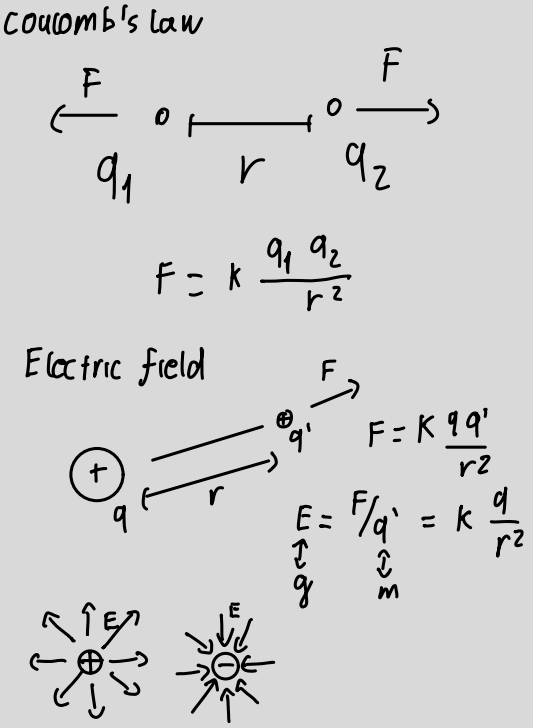
Current, Resistance

& Magnetic Field

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Today's Topics

- **Current and Resistance**
  - The Electric Potential
  - The Electron Current
  - Resistance and Ohm's Law
- **The Magnetic Field**
  - Magnetism
  - Magnetic Field of a Current
  - Magnetic Force



Electrical potential energy ⇔ gravitational potential energy

$\Delta U = (-W) \rightarrow$ kinetic energy – work theorm:

$\Delta KE = W = -\Delta U$

$\Rightarrow \underbrace{\Delta KE + \Delta U}_{\Delta E} = 0$

$\oplus_{q_1} \quad \oplus_{q_2} \longrightarrow F = k\frac{q_1 q_2}{r^2}$

Work of q_2 (Force is not constant)

$$W = \int_{x_1}^{x_2} F\,dx = \int k\frac{q_1 q_2}{x^2}\,dx \Rightarrow U = k\frac{q_1 q}{x}$$

1. The Electric Potential

Analogous to Electric Field

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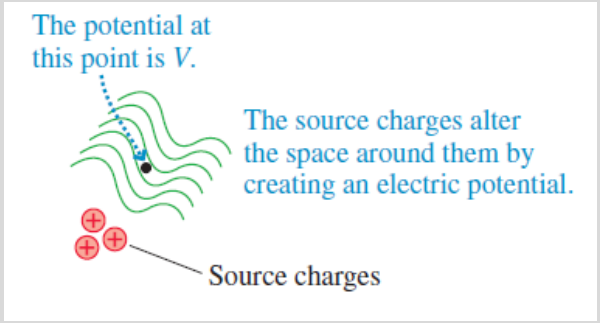
The Electric Potential

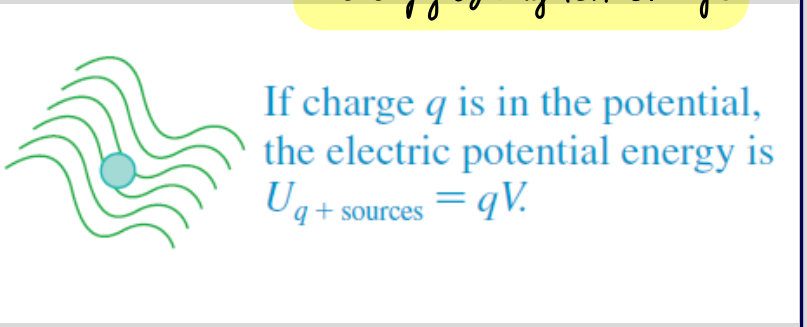
- In analogy with the electric field, we will define the **electric potential *V*** which can be quantified as:

$$V = \frac{U_{q+sources}}{q}$$

$V = k\frac{q_1}{x}$ $U = Vq_2$

Calculate the electrical potential energy of any test charge.





$J/C = V$

- The unit of electric potential is **Joule per coulomb**, which is called **Volt (*V*)**.
- The potential difference between two points is called the **voltage**. $\Delta V = V_1 - V_2$

The Electric Potential

- Recall our conservation of energy equation:

$$K_i + U_i = K_f + U_f$$

$m(gh)$ $m(gh)$

- Now we can write it in terms of the **Electric Potential** as:

$$K_i + qV_i = K_f + qV_f$$

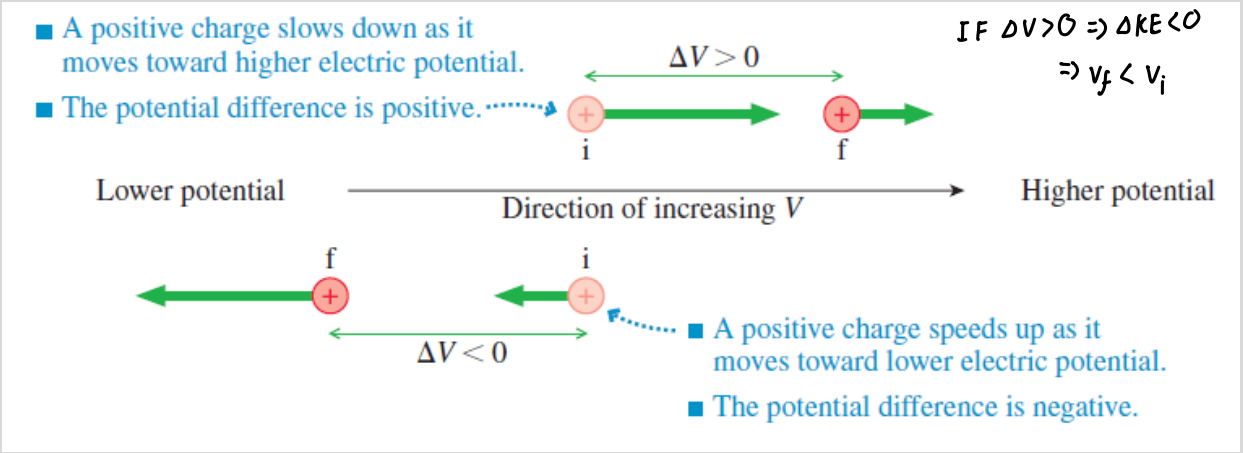
$\Delta KE + \Delta U = 0$

$\downarrow$

$q\Delta V$

IF $\Delta V > 0 \Rightarrow \Delta KE < 0$

$\Rightarrow v_f < v_i$



The Electric Potential

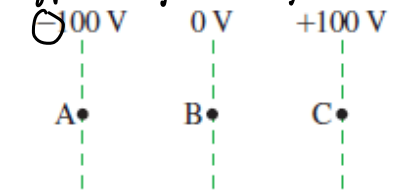
Question: A proton is released from rest at point B, where the potential is 0 V. Afterward, the proton

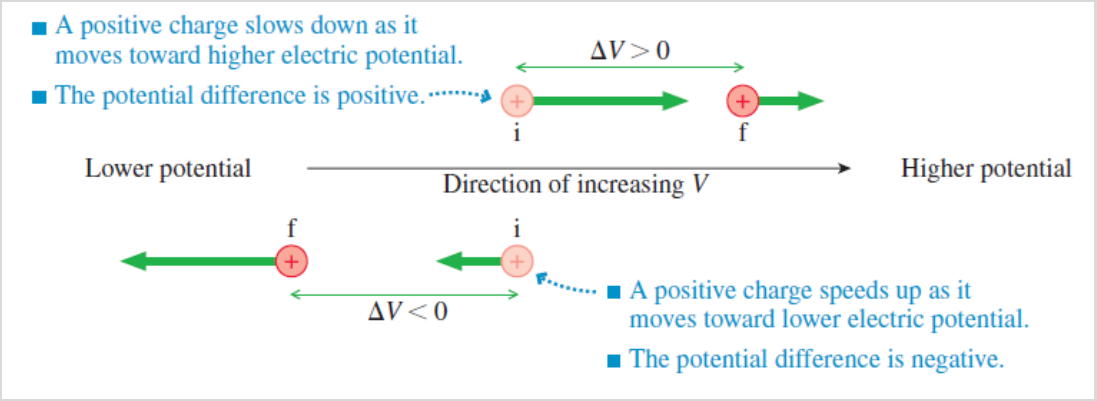
a. Remains at rest at B.
b. Moves toward A with a steady speed.
c. Moves toward A with an increasing speed.
d. Moves toward C with a steady speed.
e. Moves toward C with an increasing speed.

$\Delta KE > 0$

$\Delta V = -100\,V < 0$

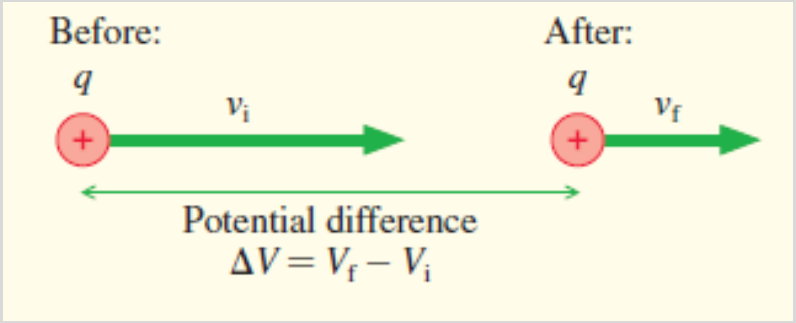
suggest a negative charge





The Electric Potential

Example: A proton (charge $e = +1.60 \times 10^{-19} C$) with a speed of $2.0 \times 10^5\ m/s$ enters a region of space with an electric potential of +100 V. What is the proton's speed after it moves through this potential difference? $m_{proton} = 1.67 \times 10^{-27} kg$



$KE_i + qV_i = KE_f + qV_f$ $\quad V_f = ?$, $V_i = 0V$, $V_f = 100V$

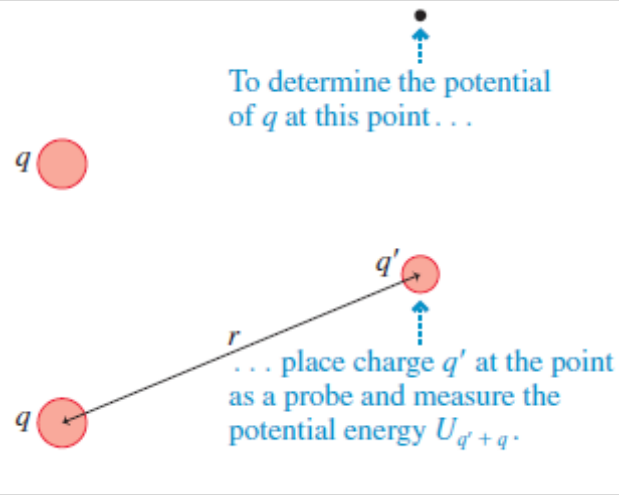
$\frac{1}{2}mv_i^2 + qV_i = \frac{1}{2}mv_f^2 + qV_f$

$\Rightarrow v_f = \sqrt{\frac{\left(\left(\frac{1}{2}mv_i^2 + qV_i\right) - qV_f\right)2}{m}}$

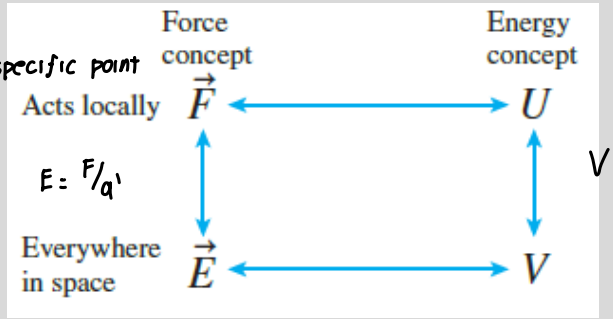
The Electric Potential
of a Point Charge

Let q be the source charge and let a second charge q' probe the electric potential of q. **The potential energy** of the two point charges is:

$$U_{q'+q} = K\frac{qq'}{r} \Rightarrow V = \frac{U_{q'+q}}{q'} = K\frac{q}{r}$$



The Electric Potential
connecting Potential and Field



only at a specific point

Relate E & V

$E = F/q'$

$V = U/q'$

- So far we have established that the **Electric Potential** is:

$$V = \frac{U_{q+sources}}{q}$$

- In addition, we defined the **electrical potential energy** to be:

$$\Delta U = -W = -\int_{s_i}^{s_f} F ds = -\int q E \, ds \Rightarrow \frac{\Delta U}{q} = -\int E \, ds$$

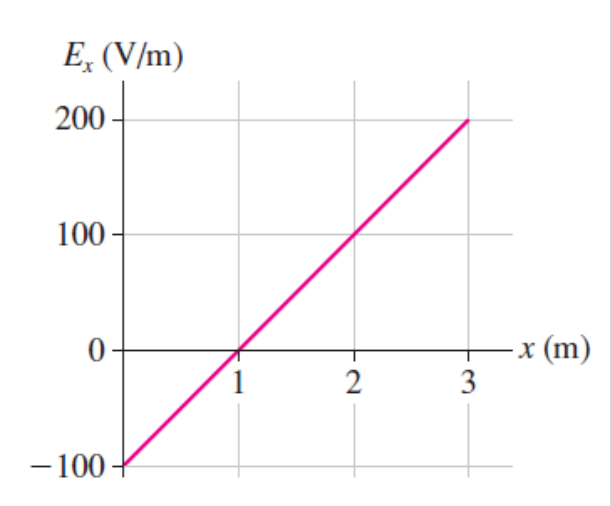
$$\Delta V = -\int E \, ds$$

- Also, we have the definition of the electric field as:

$$\vec{F} = q\vec{E}$$

The Electric Potential
connecting Potential and Field

Example: The graph below shows the electric field E_x which varies along the $x-axis$. What is the potential difference between $x_i = 1\ m$ and $x_f = 3.0\ m$?



$\Delta V = ?$ from $1m \rightarrow 3m$

$= -\int_1^3 E\,dx$ = (area under the curve)

$= -\frac{(2 \times 200)}{2} = -200\ V$

2. The Electron Current

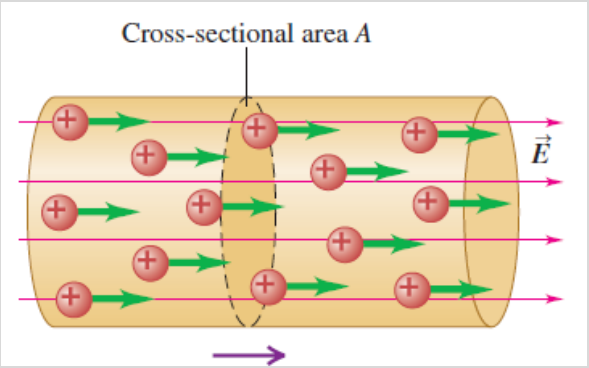
Explore the *controlled motion* of charges

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Electric Current

Current

Suppose ΔQ is the amount of charge that flows through an area A in a time interval Δt



The average current

$\frac{\Delta V}{\Delta t}$ → strength of the flow

$$\bar{I} = \frac{\Delta Q}{\Delta t}$$

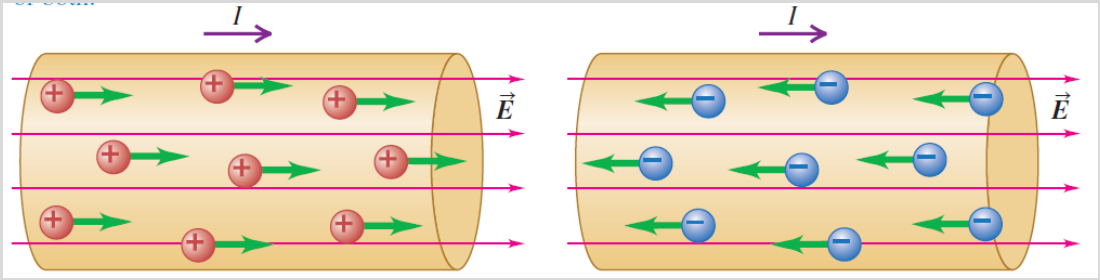
The instantaneous current

$$I = \lim_{\Delta t \to 0} \frac{\Delta Q}{\Delta t} = \frac{dQ}{dt}$$

Electric Current

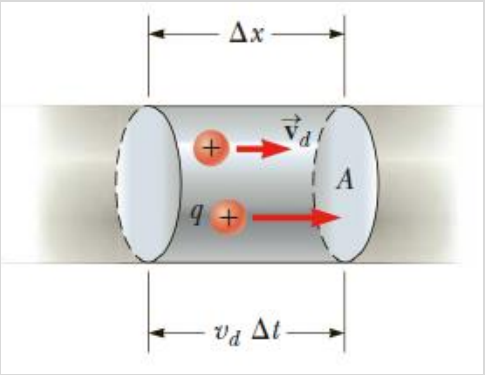
Current

A conventional current is treated as flow of POSITIVE charges



Electron Current
A microscopic view

The current depends on the average speed of the charge carriers.



The volume of an element of length Δx

$\Delta V = A\,\Delta x$

Denote n → the number of carriers per unit volume

So the number of carriers in the volume ΔV is

$n\,\Delta V$

The charge in this element $\Delta Q = (n\,\Delta V)\,q \Rightarrow I = \lim_{\Delta t \to 0} \frac{\Delta Q}{\Delta t} = nAq \underbrace{\lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t}}_{v_d}$

Then the current is

$$I = \lim_{\Delta t \to 0} \frac{\Delta Q}{\Delta t} = nqv_dA$$

Electron Current
A microscopic view

Example: What current in a copper wire with a cross-sectional area of 7.5×10^{-7} m^2 would result in a drift speed equal to 5×10^{-4} m/s? Given $q_e = 1.6 \times 10^{-19}$ C.



$I = q_e n A v_d$

$= 5.08$ A

Metal	Electron density (m^{-3})
Aluminum	6.0×10^{28}
Copper	8.5×10^{28}
Iron	8.5×10^{28}
Gold	5.9×10^{28}
Silver	5.8×10^{28}

3. Resistance and Ohm's Law

Linking Electric Potential and Current

V I

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Resistance

Examine a section of wire of length L with a potential difference between the ends

$$\Delta V = V_+ - V_-$$

Goal: relate the current I and the potential difference ΔV

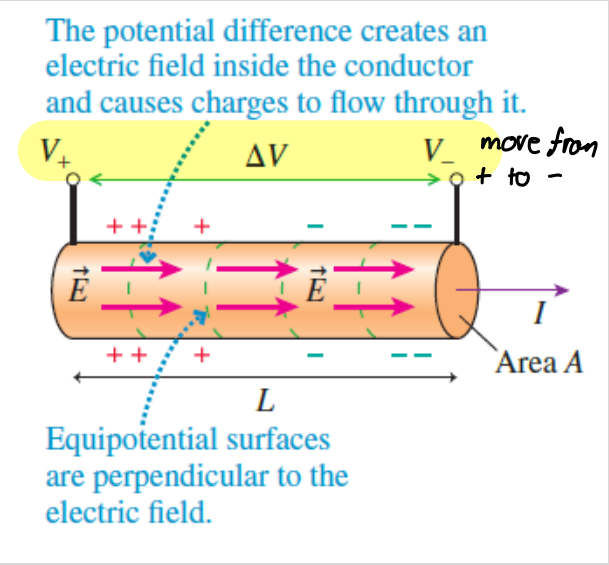
Ohm's Law states:

$$I = \frac{\Delta V}{R}$$

- R is the electrical resistance of the conductor

- The unit of electrical resistance is Ohm.

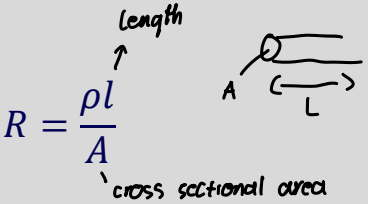
$1\ ohm = 1\Omega = 1\ V/A$



Resistance

The resistance can be computed as

$$R = \frac{\rho l}{A}$$



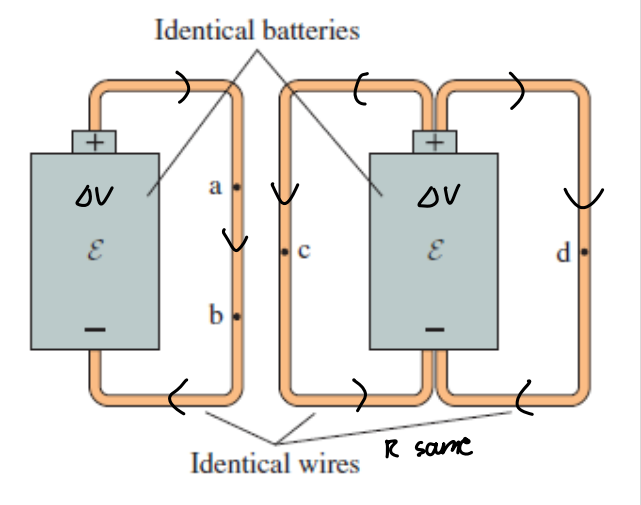
where

ρ: the resistivity of the material.

Material	Resistivity ($\Omega \cdot m$)
Silver	1.59×10^{-8}
Copper	1.7×10^{-8}
Gold	2.44×10^{-8}
Aluminum	2.82×10^{-8}
Tungsten	5.6×10^{-8}
Iron	10.0×10^{-8}
Platinum	11×10^{-8}
Lead	22×10^{-8}
Nichrome[a]	150×10^{-8}
Carbon	3.5×10^{-5}

Ohm's Law

Example: A wire connects the positive and negative terminals of a battery A. Two identical wires connect the positive and negative terminals of battery B (which is identical to battery A). Rank in order from largest to smallest, the currents I_a to I_d at points a to d:



$$I_A = I_B = \frac{\Delta V}{R}$$

$$I_C = I_D = \frac{\Delta V}{R}$$

$$\therefore I_a = I_B = I_C = I_d$$

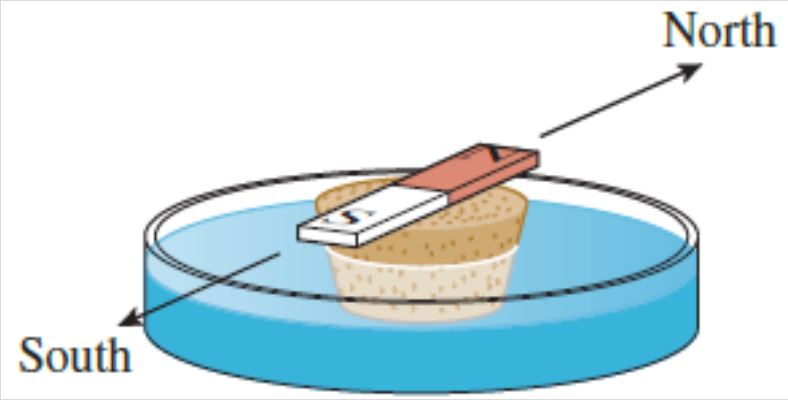
4. Magnetic Field

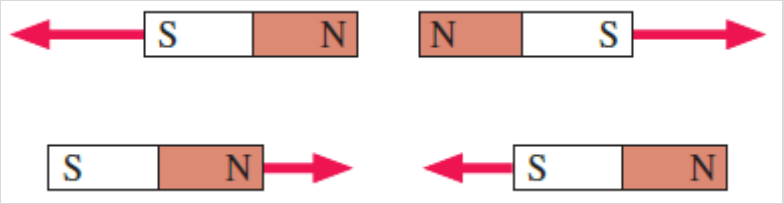
Linking Electric Potential and Current

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Magnetic Field
Magnetism

Experiments with magnets

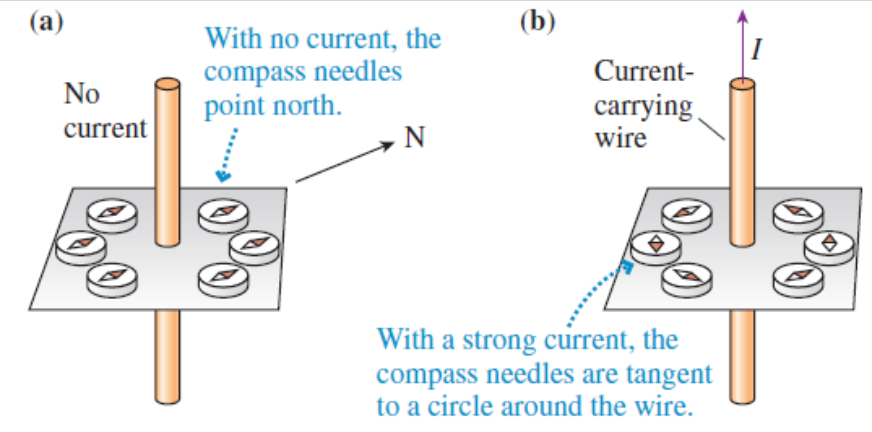




Magnetic Field

Connection between electricity and manetism

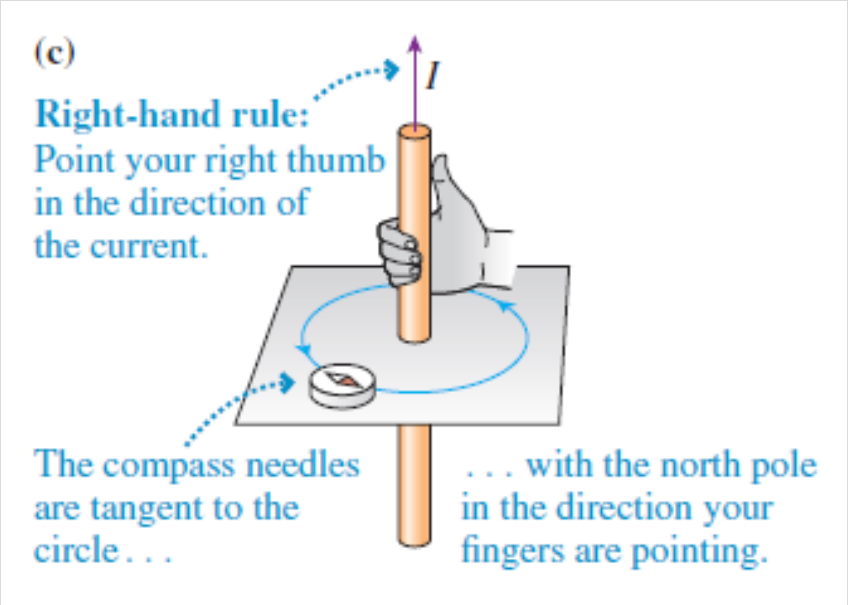
Response of compass needles to a current in a straight wire



Magnetic Field

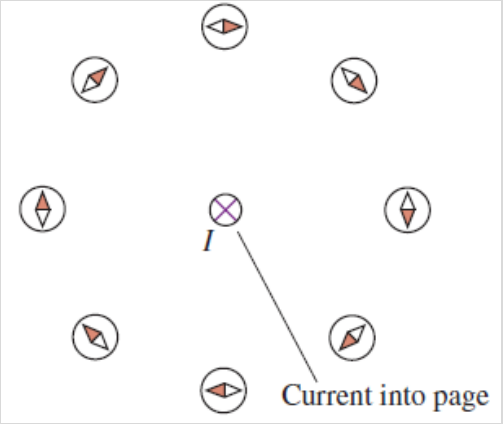
Connection between electricity and manetism

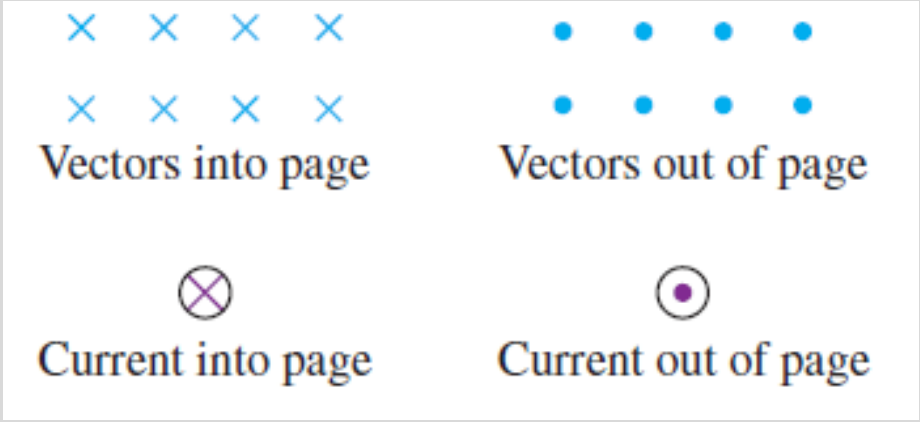
Response of compass needles to a current in a straight wire



Magnetic Field
Definition

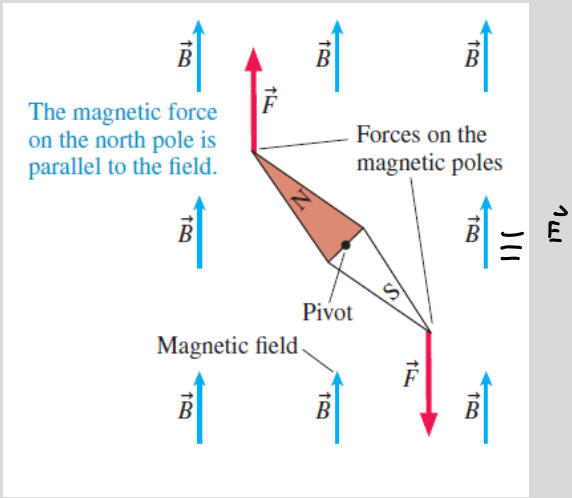
Notations in 3D space

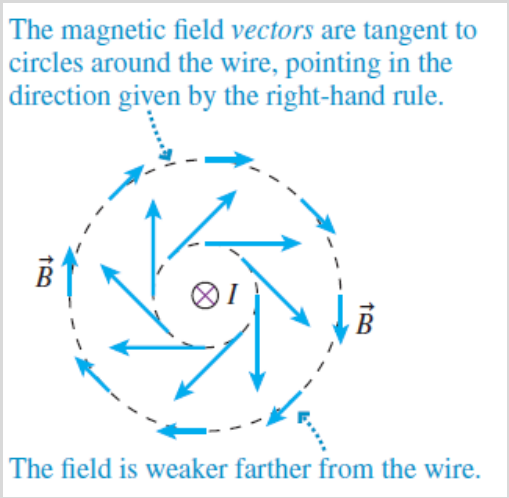




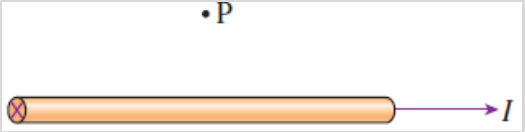
Magnetic Field
Definition

The magnetic field exerts forces on the poles of a compass.





Question: The magnetic field at position P points



(a) Up. (b) Down. (c) Into the page. (d) Out of the page

Soure of Magnetic Field

Moving Charges

need consult

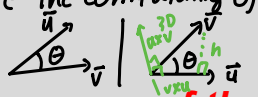
|u| |v| sin θ mean area of the Parallelogram created by u & v



Since current in a wire generates a magnetic field, and a current is a collection of moving charges

cross product → help us to facilitate the computating of the magnetic field $\vec{B}$

Dot product: $\vec{u} \cdot \vec{v} = |u||v|\cos\theta$ → a scalar

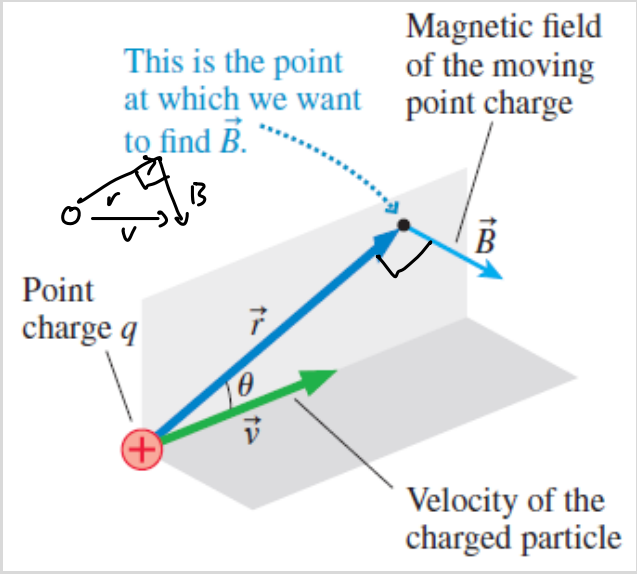


$|\vec{u} \times \vec{v}| = |\vec{u}||\vec{v}|\sin\theta$ cross

→ we can guess that ***moving charges are the source of the magnetic field***

$$\vec{B}_{point\ charge} = \frac{\mu_0}{4\pi}\frac{qv\sin(\theta)}{r^2} = \frac{\mu_0}{4\pi}\frac{q\vec{v}\times\hat{r}}{r^2}$$

(velocity; distance)



→ **Biot – Savart Law**

- Direction given by the right-hand rule.
- SI unit of B is **Tesla (T)**, it is defined as

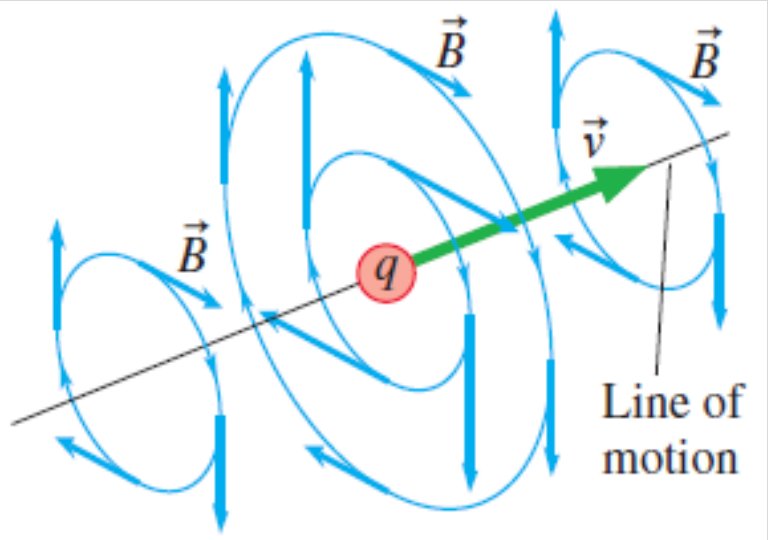
$$1\ \text{Tesla} = 1\ T = 1\frac{N}{A}m$$

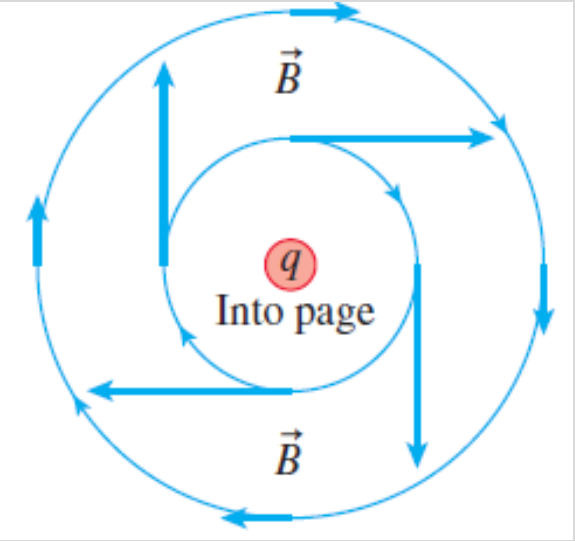
- μ_0 is the permeability constant of vacuum $4\pi \times 10^{-7}\ T\ m/A$

Soure of Magnetic Field

Moving Charges

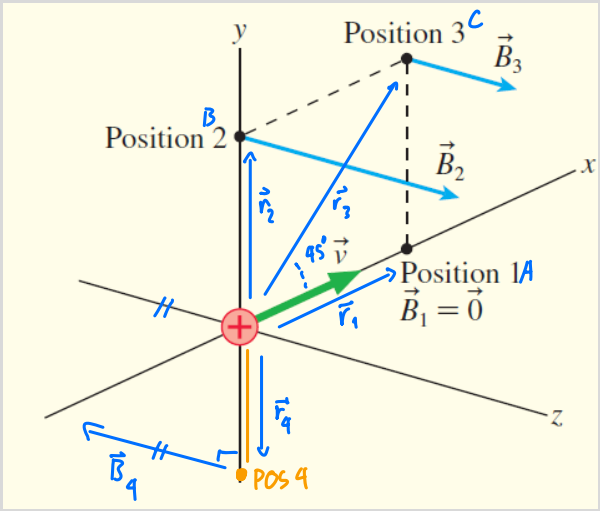
Two views (3D & 2D) of the magnetic field of a moving positive charge





Magnetic Field

Example: A proton moves with velocity $\vec{v} = 1.0 \times 10^7\ \hat{i}$ m/s. As it passes the origin, what is the magnetic field at the positions A (1, 0, 0)mm, B(0, 1, 0) mm, C(1, 1, 0) mm?



$$B = \frac{\mu_0}{4\pi}\frac{qv\sin\theta}{r^2}$$

unit vector along x, y, z → $\hat{i}, \hat{j}, \hat{k}$

At A: $\theta = 0° \Rightarrow B_1 = 0$

At B: $\theta = 90° \Rightarrow B_2 = \frac{\mu_0}{4\pi}\frac{qv}{r_2^2}\sin 90 \Rightarrow \vec{B}_2 = B_2\,\hat{k}$ ①

At C: $\theta = 45° \Rightarrow \sin\theta = \frac{\sqrt{2}}{2} \Rightarrow B_3 = \frac{\mu_0}{4\pi}\frac{qv}{r_3^2}\cdot\frac{\sqrt{2}}{2}$

$\vec{B}_q = -B_q\,\hat{k}$

Direction: $\vec{B}_2 = \frac{\mu_0}{4\pi}\frac{q\vec{v}\times\hat{r}_2}{r_2^2}$

In general $\vec{u} = u_x\hat{i} + u_y\hat{j} + u_z\hat{k}$

$\vec{v} = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$

$\Rightarrow \vec{u}\times\vec{v}$ = the determinant of a matrix

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ u_x & u_y & u_z \\ v_x & v_y & v_z \end{vmatrix}$$

$$= \hat{i}\begin{vmatrix} u_y & u_z \\ v_y & v_z \end{vmatrix} - \hat{j}\begin{vmatrix} u_x & u_z \\ v_x & v_z \end{vmatrix} + \hat{k}\begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}$$

$(u_y v_z - v_y u_z)$ $(u_x v_z - v_x u_z)$ $(u_x v_y - v_x u_y)$

Magnetic Field
Cross Product

- Similar to Electric Fields, **Magnetic Field is a vector field** which points in the direction given by the right-hand rule.
- Here, we shall use the **Vector Cross-Product** to obtain the full definition of the Biot-Savart law.
- The definition of a Cross Product between 2 vectors $\vec{u}[\mathrm{u_x}, \mathrm{u_y}, \mathrm{u_z}]$ and $\vec{v}[v_x, v_y, v_z]$ is as follows:

$$\vec{\boldsymbol{u}} \times \vec{\boldsymbol{v}} = [\boldsymbol{u_x}, \boldsymbol{u_y}, \boldsymbol{u_z}] \times [\boldsymbol{v_x}, \boldsymbol{v_y}, \boldsymbol{v_z}]$$

$$= [(\boldsymbol{u_y} \cdot \boldsymbol{v_z} - \boldsymbol{u_z} \cdot \boldsymbol{v_y}), (\boldsymbol{u_z} \cdot \boldsymbol{v_x} - \boldsymbol{u_x} \cdot \boldsymbol{v_z}), (\boldsymbol{u_x} \cdot \boldsymbol{v_y} - \boldsymbol{u_y} \cdot \boldsymbol{u_x})]$$

Magnetic Field
Cross Product

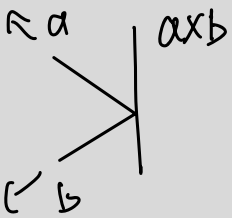
- By definition, the magnitude of the Cross Product between 2 vectors is given by:

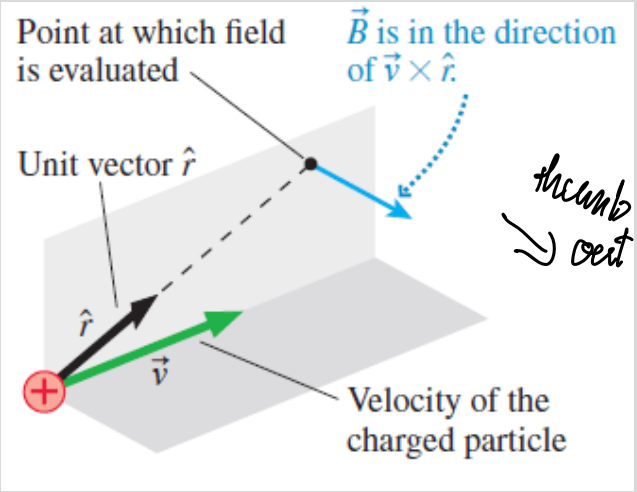
$$|\vec{u} \times \vec{v}| = |\vec{u}||\vec{v}|\sin(\theta)$$

  - Where θ is the angle between the 2 vectors

- A compact version of Biot-Savart law:

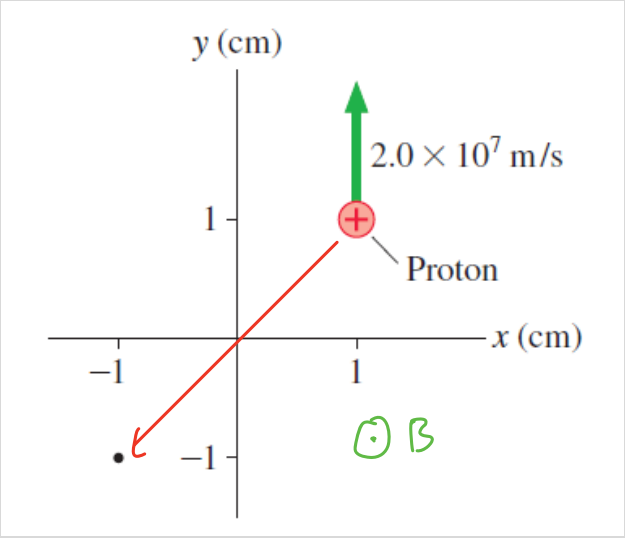
$$\vec{B}_{point\ charge} = \frac{\mu_0}{4\pi} \frac{qv\sin(\theta)}{r^2}$$





Magnetic Field

Example: What is the magnetic field at the position of the dot (-1, -1) in the figure below? Give your answer as a vector.



thumb : $\vec{B}$

1 finger : $\vec{v}$

middle : $\vec{r}$

$$B = \frac{\mu_0}{4\pi} q \frac{v}{r^2} \sin\theta$$

$\theta = 135^\circ$

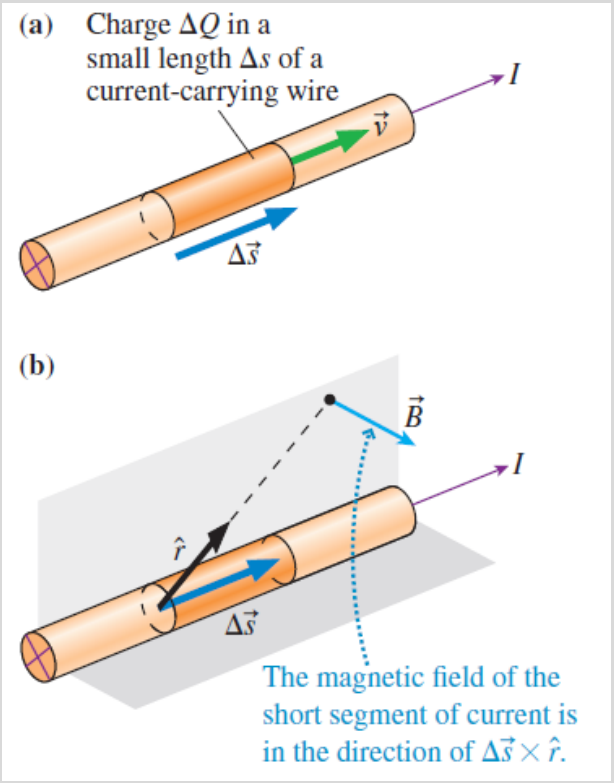
5. Magnetic Field of a Current

Magnetic field by infinite charges

RMIT UNIVERSITY

Magnetic Field of a Current

Goal: write the Biot-Savart law in terms of current.



$$\vec{B} = \frac{\mu_0}{4\pi} \frac{(\Delta Q \vec{v}) \times \hat{r}}{r^2}$$

cross

$$\Delta Q \cdot \vec{v} = \Delta Q \cdot \frac{\Delta \vec{s}}{\Delta t} = I \Delta \vec{s}$$

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{I \Delta \vec{s} \times \vec{r}}{r^2}$$

Magnetic Field of a Current

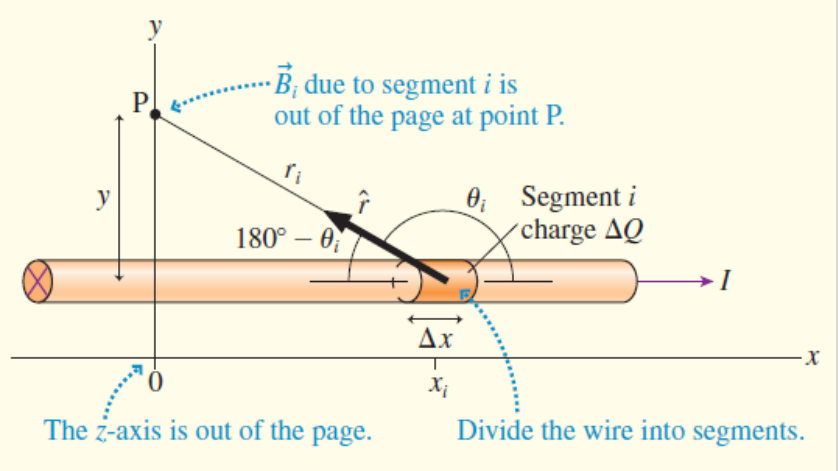
Consider an infinitely long, straight wire carries current I, find the magnetic field at distance y from the wire:

$$\overrightarrow{B_i} = \frac{\mu_0}{4\pi}\frac{I\,\Delta x\sin(\theta_i)}{r_i^2} = \frac{\mu_0}{4\pi}\frac{I\ \sin(\theta_i)}{(x_i^2+y^2)}\Delta x$$

Express $\sin(\theta_i)$ in terms of x_i *and* y

$$\sin(\theta_i) = \frac{y}{r_i} = \frac{y}{\sqrt{x_i^2+y^2}}$$

$$\overrightarrow{B_i} = \frac{\mu_0}{4\pi}\frac{I\,y}{\left(x_i^2+y^2\right)^{\frac{3}{2}}}\Delta x$$



Magnetic Field of a Current

- Note that the equation above is only applicable to a small segment of the wire. If we want to have the magnetic field of the whole wire, we should find the sum:

$$B_{wire} = \frac{\mu_0 I y}{4\pi} \sum_i \frac{\Delta x}{\left(x_i^2 + y^2\right)^{\frac{3}{2}}} = \frac{\mu_0 I y}{4\pi} \int_{-\infty}^{\infty} \frac{\mathrm{d}x}{\left(x_i^2 + y^2\right)^{\frac{3}{2}}}$$

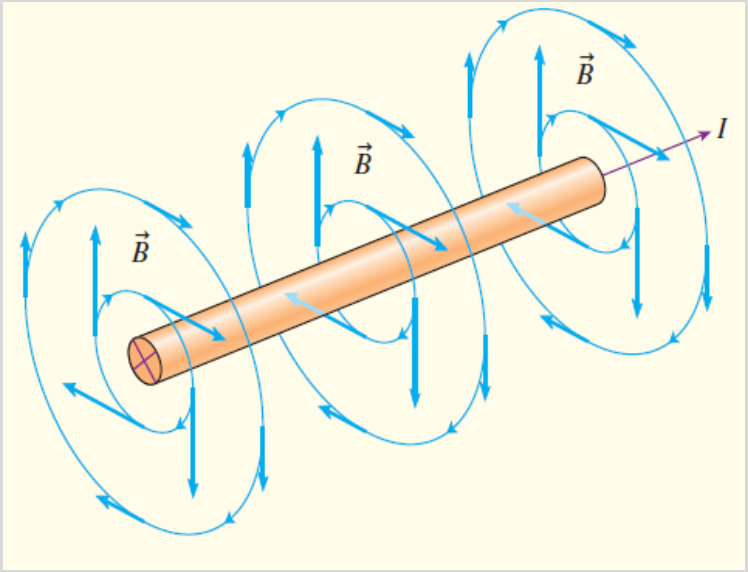
- Solving the Integration above we will obtain:

$$B_{wire} = \frac{\mu_0 I y}{4\pi} \frac{x}{y^2 (x^2 + y^2)^{\frac{1}{2}}} \Big|_{-\infty}^{\infty} = \frac{\mu_0 I}{2\pi y}$$

Magnetic Field of a Current

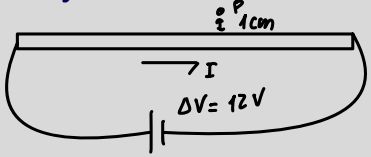
Hence, we can conclude that the magnetic field for a wire is:

$$\overrightarrow{B_{wire}} = \left(\frac{\mu_0 \, I}{2\pi \, r}, tangent\ to\ a\ circle\ around\ the\ wire \right)$$



Magnetic Field of a Current

Example: A 1.0 meter long, 1.0 mm diameter nichrome heater wire is connected to a 12 V battery. What is the magnetic field strength 1.0 cm away from the wire? Assume the resistivity of nichrome is: $\rho = 1.50 \times 10^{-6}\ \Omega m$.



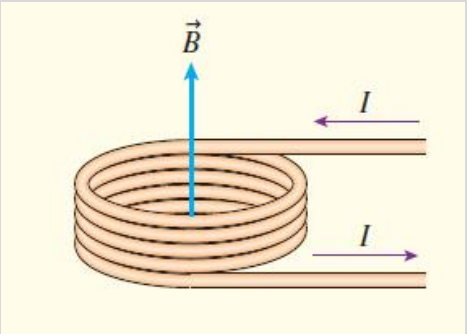
$I = \Delta V / R \qquad R = \frac{\rho L}{A} = 1.91\,\Omega$

Magnetic Field of a Current

- If the wire is bend into a loop, the equation above becomes (proof available upon request):

$$\vec{B}_{Coil\ Center} = \frac{\mu_0}{2}\frac{NI}{R}$$

  - R is the radius of the loop.
  - N is the number of turns in the loop.



Magnetic Field of a Current

Example: What current is needed in a 5-turn, 10-cm diameter coil to cancel the Earth's magnetic field at the center of the coil? Given that the Earth's magnetic field is about $5 \times 10^{-5}\ T$.

